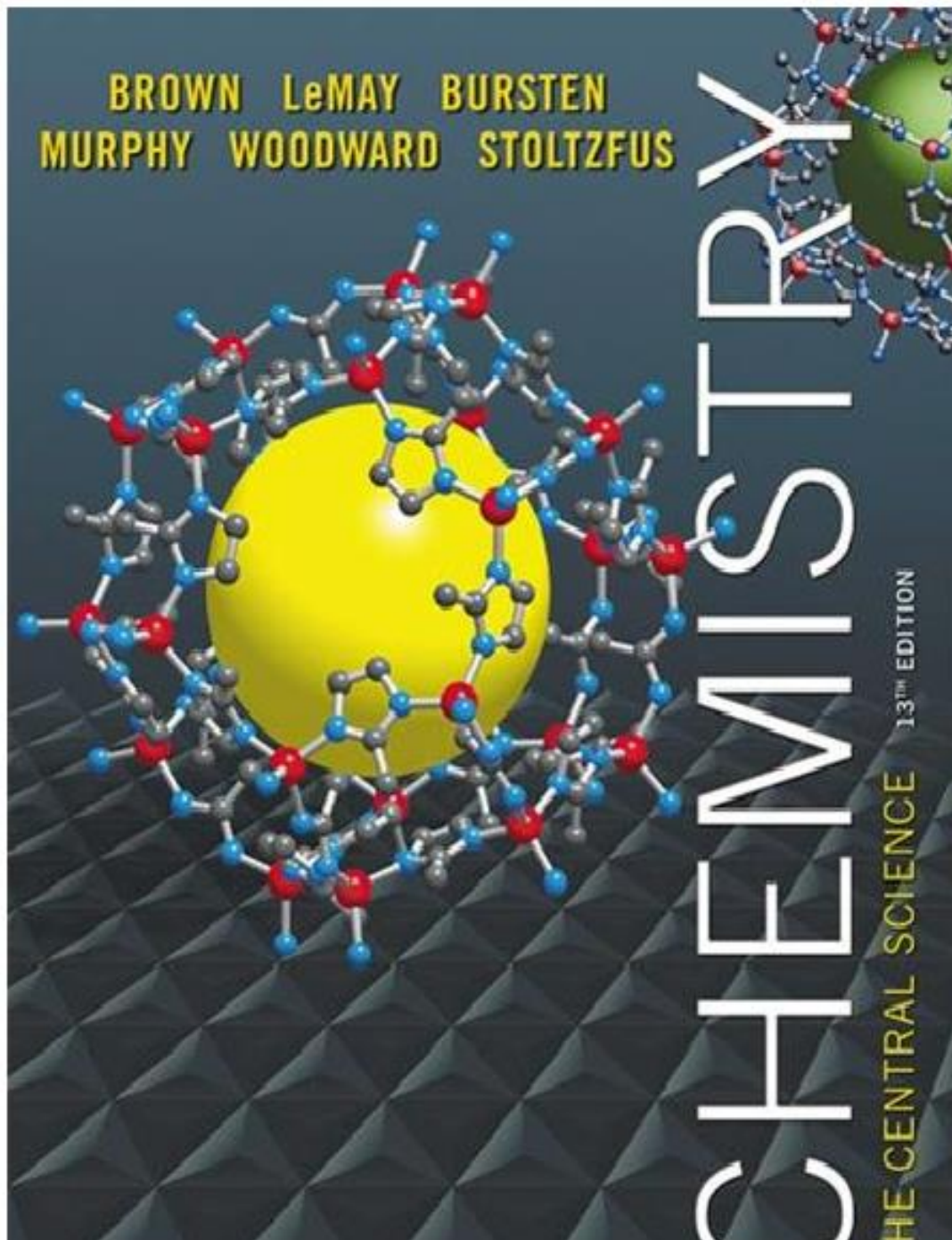




Additional Aspects of Aqueous Equilibria

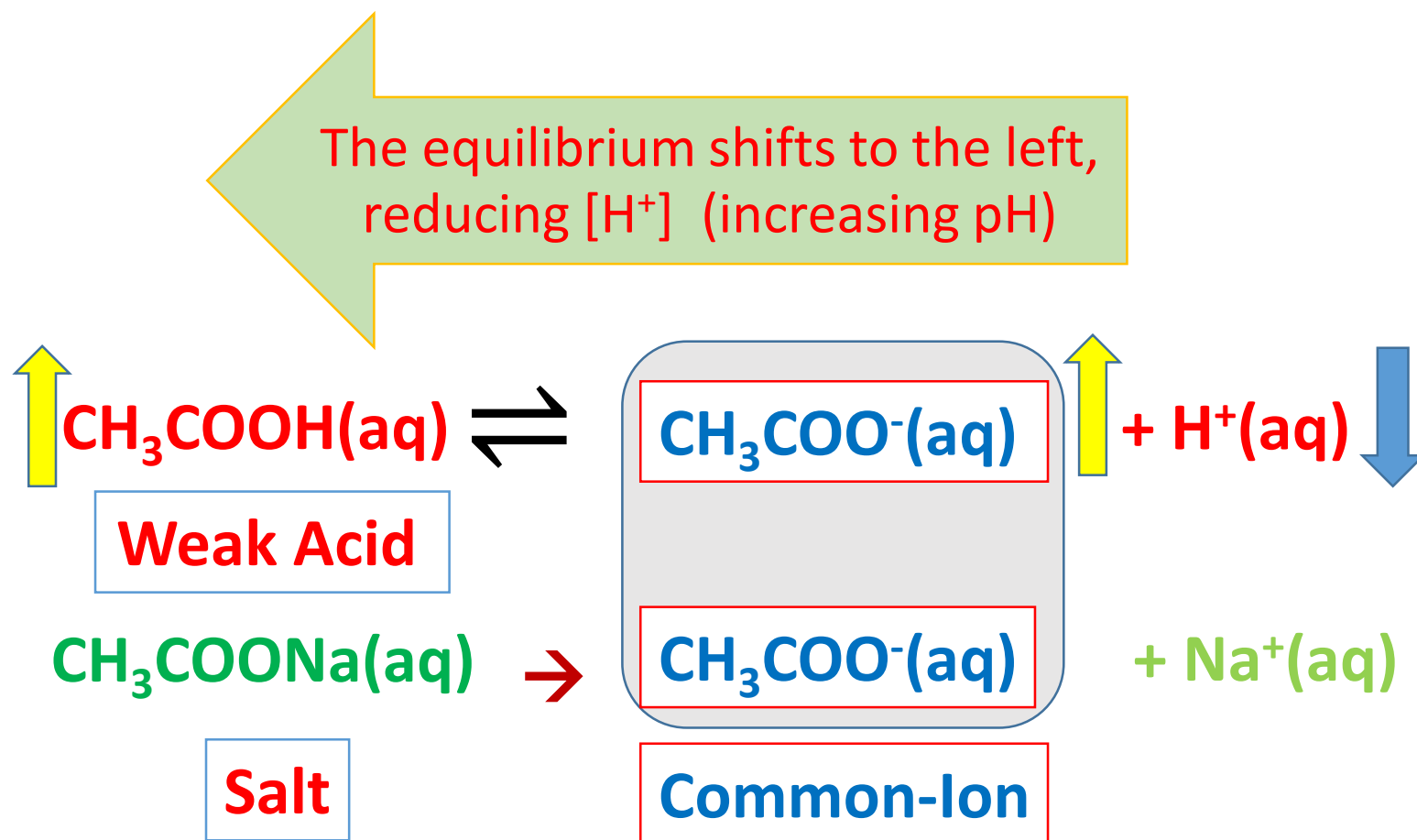
17.1 The Common-Ion Effect

17.2 Buffers

17.1 THE COMMON-ION EFFECT

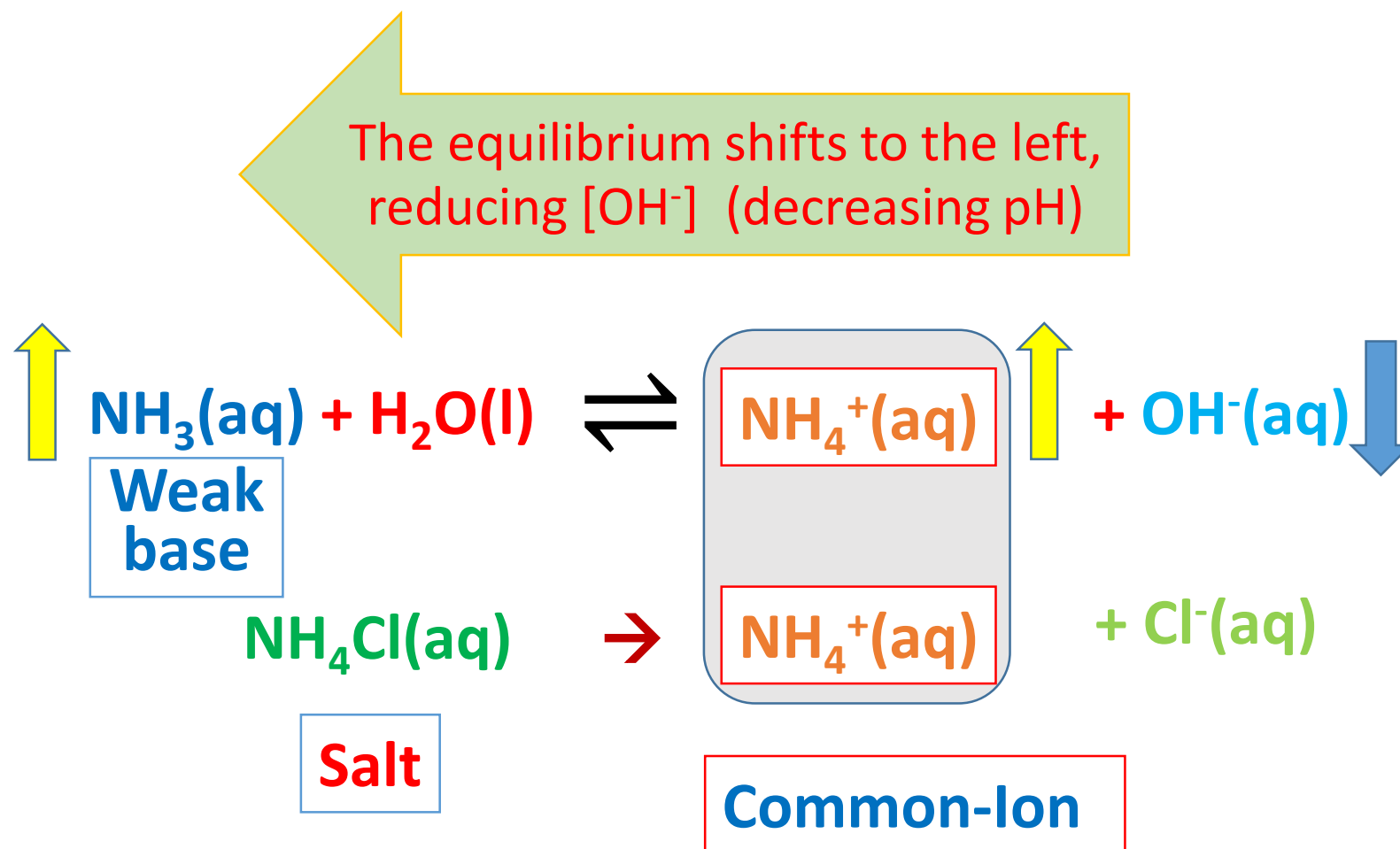
Effect of Acetate on the Acetic Acid Equilibrium

■ **The Common-Ion Effect:** Whenever a weak electrolyte and a strong electrolyte containing a common ion are together in solution, the weak electrolyte ionizes less than it would if it were alone in solution."



17.1 THE COMMON-ION EFFECT

Effect of ammonium on the ammonia equilibrium



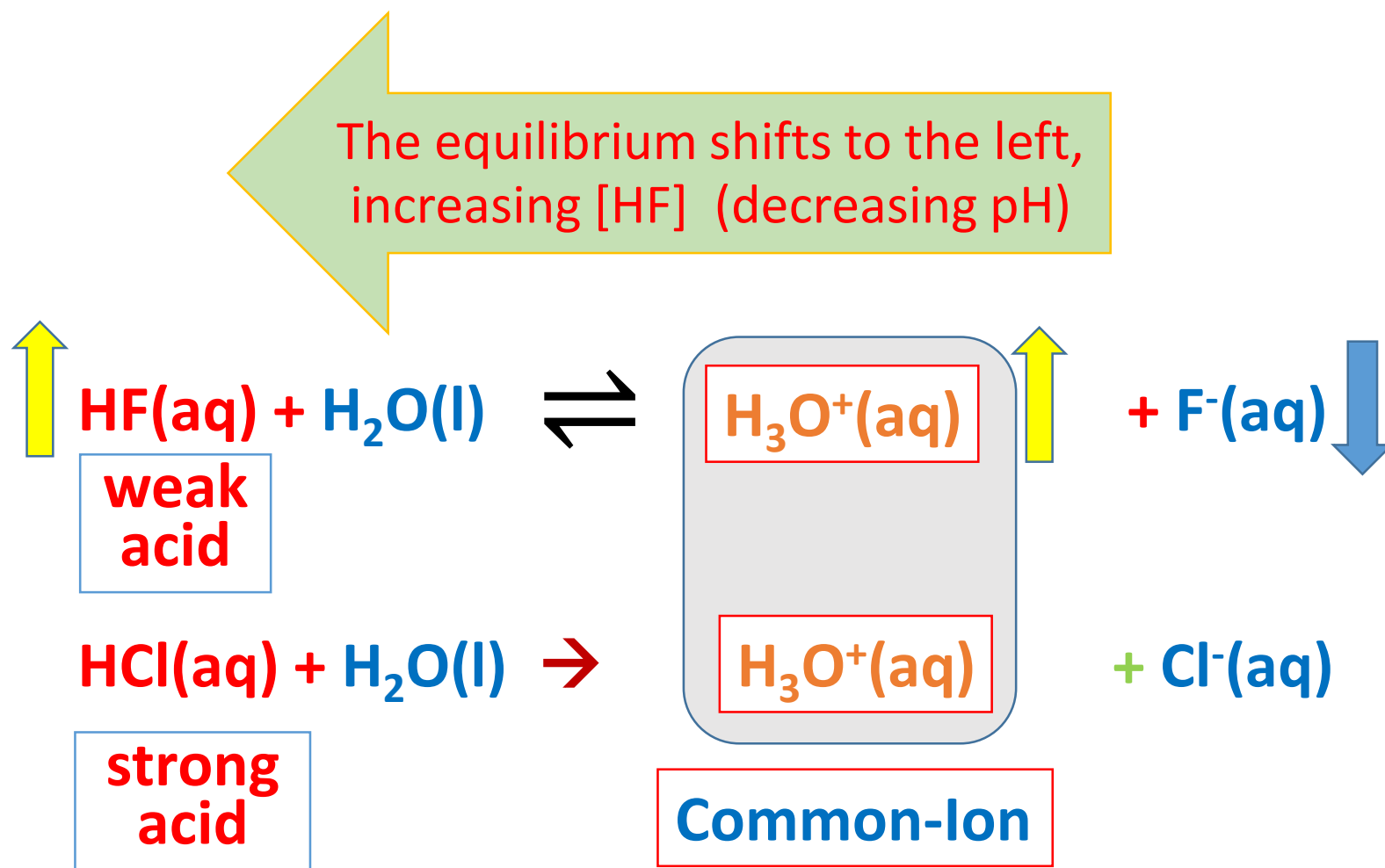
Sample Exercise: Calculate the $[F^-]$ and pH of a solution that is 0.20 M in HF . K_a for HF is 6.8×10^{-4} .

	$HF(aq) + H_2O(l)$	$\rightleftharpoons$	$H_3O^+(aq) + F^-(aq)$
Initial:	0.20		0.00 0.00
Change:	- x		+ x + x
Equilibrium:	0.20 - x		x x

$$K_a = \frac{[H_3O^+][F^-]}{[HF]} = 6.8 \times 10^{-4} \implies \frac{x^2}{0.20 - x} = 6.8 \times 10^{-4}$$

$$x = 1.17 \times 10^{-2} = [F^-] = [H_3O^+] \implies pH = 1.93$$

Sample Exercise: Calculate the $[F^-]$ and pH of a solution that is 0.20 M in HF and 0.10 M in HCl . K_a for HF is 6.8×10^{-4} .



Sample Exercise: Calculate the $[F^-]$ and **pH** of a solution that is 0.20 *M* in HF and 0.10 *M* in HCl. K_a for HF is 6.8×10^{-4} .

	$HF(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + F^-(aq)$		
Initial:	0.20	0.10	0.00
Change:	- x	+ x	+ x
Equilibrium:	0.20 - x	0.1 + x	x

$$K_a = \frac{[H_3O^+][F^-]}{[HF]} = 6.8 \times 10^{-4}$$

$$\frac{(0.1 + x)(x)}{(0.20 - x)} = 6.8 \times 10^{-4}$$

$$\frac{(0.1)(x)}{(0.20)} = 6.8 \times 10^{-4}$$

$$\Rightarrow x = 1.36 \times 10^{-3} = [F^-]$$

$$\text{pH} = 0.98$$

$$0.10 + 1.36 \times 10^{-3} = [H_3O^+]$$

$$0.1036 \text{ M} =$$

17.2 Buffers



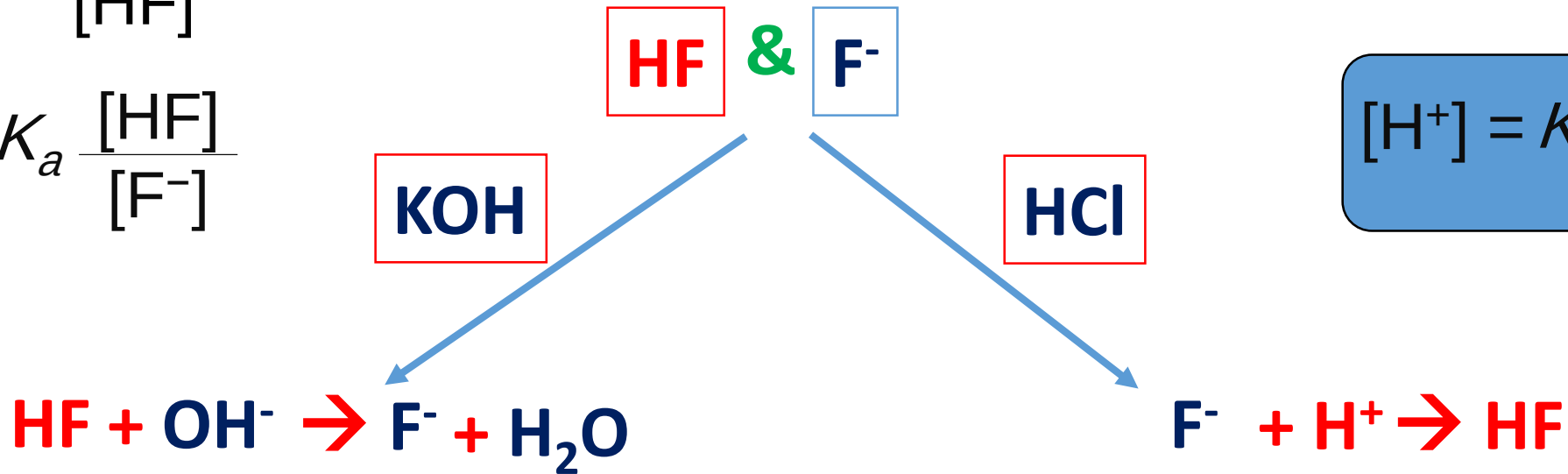
- A buffer is a solution that contains comparable amounts of a weak acid and its conjugate base $\{\text{CH}_3\text{CO}_2\text{H} \text{ \& \; } \text{CH}_3\text{CO}_2^-\}$, or a weak base and its conjugate $\{\text{NH}_3 \text{ \& \; } \text{NH}_4^+\}$.
- They resist drastic changes in pH when small amounts of strong acid or base are added.

$$K_a = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]}$$



$$[\text{H}^+] = K_a \frac{[\text{HF}]}{[\text{F}^-]}$$

$$[\text{H}^+] = K_a \frac{[\text{acid}]}{[\text{base}]}$$

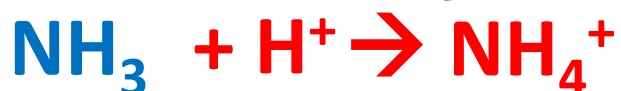


This causes [HF] to decrease & [F⁻] to increase. As long as the amounts of HF & F⁻ in the buffer are large relative to the amounts of OH⁻ added, the ratio of [HF]/[F⁻] does not change much and, thus the change in pH is small.

This causes [HF] to increase & [F⁻] to decrease. As long as the amounts of HF & F⁻ in the buffer are large relative to the amounts of H⁺ added, the ratio of [HF]/[F⁻] does not change much and, thus the change in pH is small.

$$K_b = \frac{[\text{OH}^-][\text{NH}_4^+]}{[\text{NH}_3]}$$

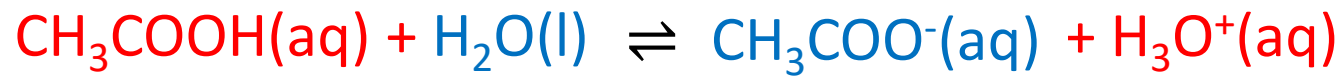
$$[\text{OH}^-] = K_b \frac{[\text{NH}_3]}{[\text{NH}_4^+]}$$



$$[\text{OH}^-] = K_b \frac{[\text{base}]}{[\text{acid}]}$$

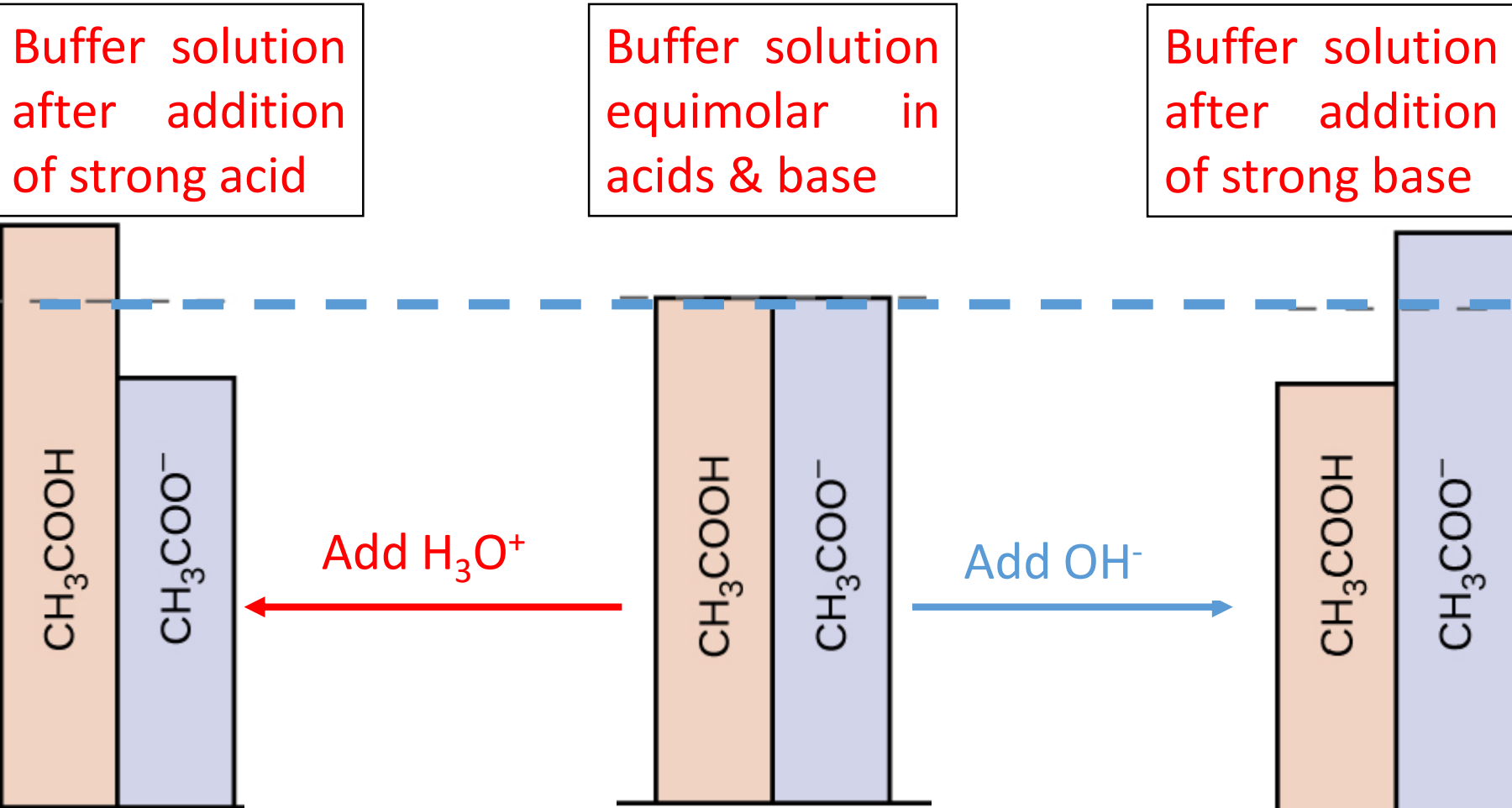
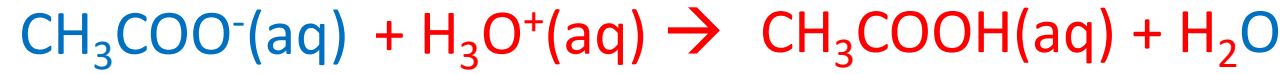
This causes $[\text{NH}_3]$ to decrease & $[\text{NH}_4^+]$ to increase. As long as the amounts of NH_3 & NH_4^+ in the buffer are large relative to the amounts of H^+ added, the ratio of $[\text{NH}_4^+]/[\text{NH}_3]$ does not change much and, thus the change in pH is small.

This causes $[\text{NH}_3]$ to increase & $[\text{NH}_4^+]$ to decrease. As long as the amounts of HF & F^- in the buffer are large relative to the amounts of OH^- added, the ratio of $[\text{NH}_4^+]/[\text{NH}_3]$ does not change much and, thus the change in pH is small.



H_3O^+ added, equilibrium shifts to left

OH^- added, equilibrium shifts to right



Calculating the pH of a Buffer



$$[\text{H}^+] = K_a \frac{[\text{acid}]}{[\text{base}]}$$



$$[\text{OH}^-] = K_b \frac{[\text{base}]}{[\text{acid}]}$$

- Taking the negative log of both side, we get

$$-\log[\text{H}^+] = -\log K_a + -\log \frac{[\text{acid}]}{[\text{base}]}$$

$$\log[\text{OH}^-] = -\log K_b + -\log \frac{[\text{base}]}{[\text{acid}]}$$

$$\text{pH} = \text{pK}_a + \log \frac{[\text{base}]}{[\text{acid}]}$$

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{acid}]}{[\text{base}]}$$

Henderson–Hasselbalch
equation

- When $[\text{base}] = [\text{acid}]$, we have $\text{pH} = \text{pK}_a$ and $\text{pOH} = \text{pK}_b$.

Sample Exercise: What is the pH of a buffer that is 0.12 M in lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and 0.10 M in sodium lactate, $\text{CH}_3\text{CH}(\text{OH})\text{COONa}$? K_a for lactic acid is 1.4×10^{-4} .

$$\text{pH} = \text{pK}_a + \log \frac{[\text{base}]}{[\text{acid}]}$$

$$= -\log (1.4 \times 10^{-4}) + \log \frac{(0.10)}{(0.12)}$$

$$= 3.85 + (-0.08)$$

$$= 3.77$$

Buffer Capacity

- **Buffer Capacity**: is the amount of acid or base the buffer can neutralize before the pH begins to change to an appreciable degree.
- The greater the amount of acid and base making the buffer the greater the Buffer Capacity.

Sample Exercise: calculate the pH of a 1.0 L solution: (K_a CH_3COOH is 1.4×10^{-4})

- a) that is 1.0 M in CH_3COOH & 1.0 M in CH_3COONa .
- b) that is 0.10 M in CH_3COOH & 1.00 M in CH_3COONa .

Since the $[\text{acid}] = [\text{base}]$, for both solutions a & b, the $\text{pH} = \text{p}K_a = 4.74$

c) Which solution has higher buffer capacity?

Solution **(a)** has greater buffer capacity than solution **(b)**, because it contains more CH_3COOH and CH_3COO^- .

pH Range

- The **pH range** is the range of pH values over which a buffer system works effectively.

Buffers most effectively resist a change in pH in either direction



$$pK_a = \text{pH}$$

$$[\text{weak acid}] = [\text{base}]$$

- When $[\text{HA}]$ and $[\text{A}^-]$ differ by more than a factor of 10, the buffer has lost its ability to absorb acid/base.

$$10 \geq \frac{[\text{conjugate base}]}{[\text{weak acid}]} \geq 0.1$$

$$\text{pH} = \text{pK}_a \pm 1$$

Addition of Strong Acids or Bases to Buffers

strong acid reacts completely with conjugate base component of buffer



$$[HX] = [HX]_i + [H_3O^+]$$

$$[X^-] = [X^-]_i - [H_3O^+]$$

$$pH = pK_a + \log \frac{[base]}{[acid]}$$

Add strong acid

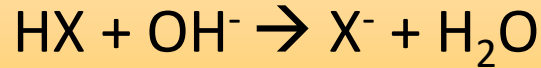
Buffer containing HX & X⁻

Calculate new values of [HX] & [X⁻]

Use K_a, [HX] & [X⁻] to calculate [H⁺]

pH

Add strong base



$$[HX] = [HX]_i - [OH^-]$$

$$[X^-] = [X^-]_i + [OH^-]$$

strong base reacts completely with weak acid component of buffer

$$pH = pK_a + \log \frac{[X^-]_i + [OH^-]}{[HX]_i - [OH^-]}$$

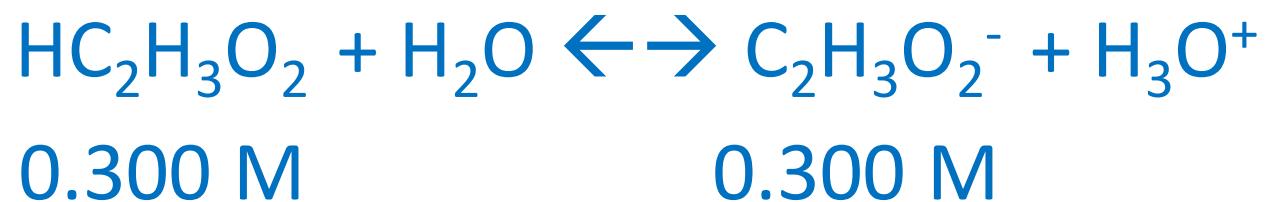
Stoichiometry calculation

Equilibrium calculation

Sample Exercise: A buffer is made by adding 0.300 mol $\text{HC}_2\text{H}_3\text{O}_2$ and 0.300 mol $\text{NaC}_2\text{H}_3\text{O}_2$ to enough water to make 1.00 L of solution.

A. Calculate the pH of the buffer.

B. Calculate the pH of this solution after 0.020 mol of NaOH is added.



$$\text{pH} = \text{pK}_a + \log \frac{[\text{base}]}{[\text{acid}]}$$



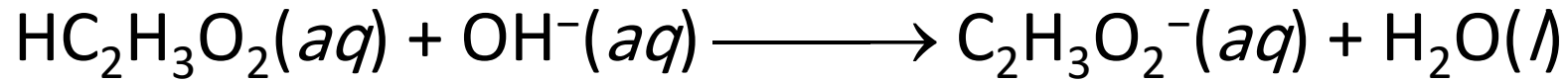
$$\text{pH} = \text{pK}_a + \log \frac{[\text{C}_2\text{H}_3\text{O}_2^-]}{[\text{HC}_2\text{H}_3\text{O}_2]}$$

$$\text{pH} = -\log(1.8 \times 10^{-5}) + \log \frac{0.30/1.0}{0.30/1.0}$$

$$\text{pH} = 4.74.$$

B. Calculate the pH of this solution after 0.020 mol of NaOH is added.

- The 0.020 mol NaOH will react with 0.020 mol of the acetic acid:

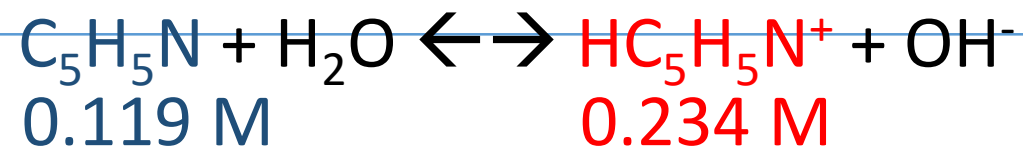


	$\text{HC}_2\text{H}_3\text{O}_2$	$\text{C}_2\text{H}_3\text{O}_2^-$	OH^-
Before reaction	0.300 mol	0.300 mol	0.020 mol
Change	- 0.020 mol	+ 0.020 mol	- 0.020 mol
After reaction	0.280 mol	0.320 mol	0.000 mol

$$\text{pH} = \text{pK}_a + \log \frac{[\text{X}^-] + [\text{OH}^-]}{[\text{HX}] - [\text{OH}^-]} \longrightarrow \text{pH} = 4.74 + \log \frac{(0.320)}{(0.280)}$$
$$= 4.74 + 0.06$$

$$\boxed{= 4.80}$$

Sample Exercise: Determine the pH of a solution that contains 0.119 M pyridine, C_5H_5N , ($pK_b = 8.77$) and 0.234 M pyridine hydrochloride, C_5H_5NCl , at 25 °C.



$$pOH = pK_b + \log \frac{[\text{acid}]}{[\text{base}]}$$



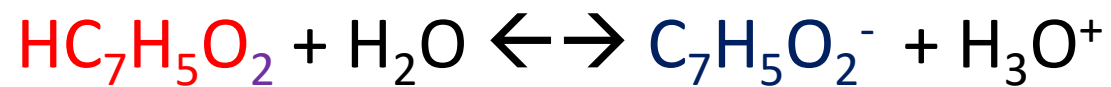
$$pOH = 8.77 + \log \frac{[HC_5H_5N^+]}{[C_5H_5N]}$$

$$pOH = 8.77 + \log \frac{[0.234]}{[0.119]}$$

$$pOH = 9.06$$

$$pH = 14 - 9.06 = 4.94$$

Sample Exercise: Calculate the concentration of sodium benzoate, $\text{NaC}_7\text{H}_5\text{O}_2$, that must be present in a 0.20 M solution of benzoic acid, $\text{HC}_7\text{H}_5\text{O}_2$ ($K_a = 6.50 \times 10^{-5}$), to produce a pH of 4.0.



0.20 M

?? M

pH = 4.0

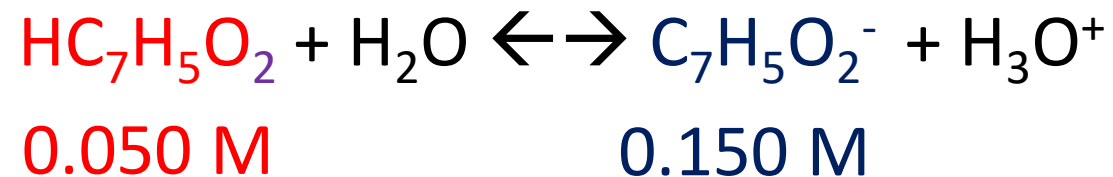
$$\text{pH} = \text{p}K_a + \log \frac{[\text{base}]}{[\text{acid}]}$$



$$4.0 = -\log(6.50 \times 10^{-5}) + \log \frac{[\text{C}_7\text{H}_5\text{O}_2^-]}{[0.2]}$$

$$[\text{C}_7\text{H}_5\text{O}_2^-] = 0.13 \text{ M}$$

Sample Exercise: Calculate the pH for a solution containing 0.0500 M benzoic acid, $\text{HC}_7\text{H}_5\text{O}_2$ ($K_a = 6.50 \times 10^{-5}$), and 0.150 M in sodium benzoate, $\text{NaC}_7\text{H}_5\text{O}_2$, at 25 °C.



$$\boxed{\text{pH} = \text{p}K_a + \log \frac{[\text{base}]}{[\text{acid}]}} \quad \longrightarrow \quad \text{pH} = -\log(6.50 \times 10^{-5}) + \log \frac{[\text{C}_7\text{H}_5\text{O}_2^-]}{[\text{HC}_7\text{H}_5\text{O}_2]}$$

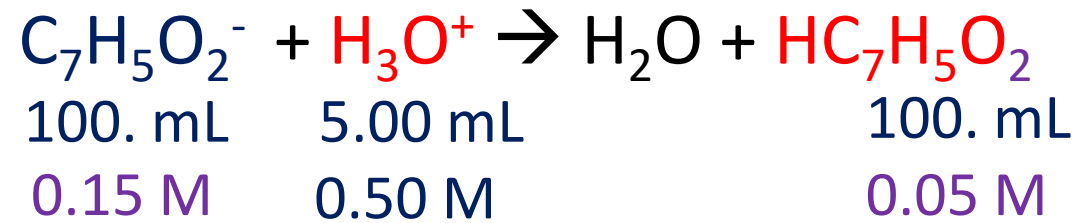
$$\text{pH} = -\log(6.50 \times 10^{-5}) + \log \frac{[0.150]}{[0.050]}$$

$$\text{pH} = 4.187 + 0.477$$

$$\boxed{\text{pH} = 4.66}$$

B. Calculate the pH when 5.00 mL of 0.50 M HCl is added to 100.00 mL of this buffered solution

- The added HCl (H_3O^+) will react with the $\text{C}_7\text{H}_5\text{O}_2^-$ conjugate base:



Mole before reaction = M x V

0.015 mol 0.0025 mol 0.005 mol

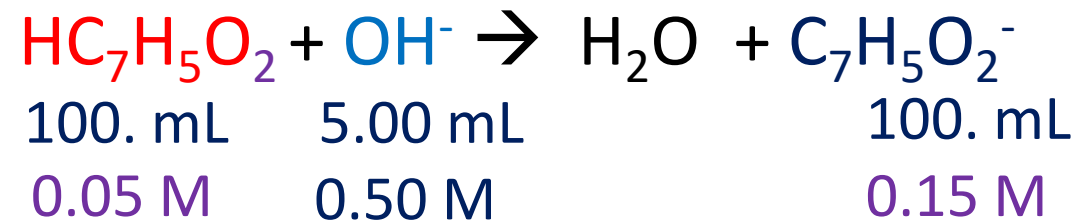
	$\text{HC}_7\text{H}_5\text{O}_2$	$\text{C}_7\text{H}_5\text{O}_2^-$	H_3O^+
Before reaction	0.0050 mol	0.015 mol	0.0025 mol
Change	+ 0.0025 mol	- 0.0025 mol	- 0.0025 mol
After reaction	0.0075 mol	0.0125 mol	0.000 mol

$$\text{pH} = \text{pK}_a + \log \frac{[\text{C}_7\text{H}_5\text{O}_2^-] - [\text{H}_3\text{O}^+]}{[\text{HC}_7\text{H}_5\text{O}_2] + [\text{H}_3\text{O}^+]} \quad \rightarrow \quad \text{pH} = 4.19 + \log \frac{(0.0125)}{(0.0075)}$$

$$\text{pH} = 4.19 + 0.22 = 4.41$$

C. Calculate the pH when 5.00 mL of 0.50 M NaOH is added to 100.00 mL of this buffered solution

- The added NaOH (OH^-) will react with the $\text{HC}_7\text{H}_5\text{O}_2$ acid:



Mole before reaction = M x V

0.005 mol 0.0025 mol

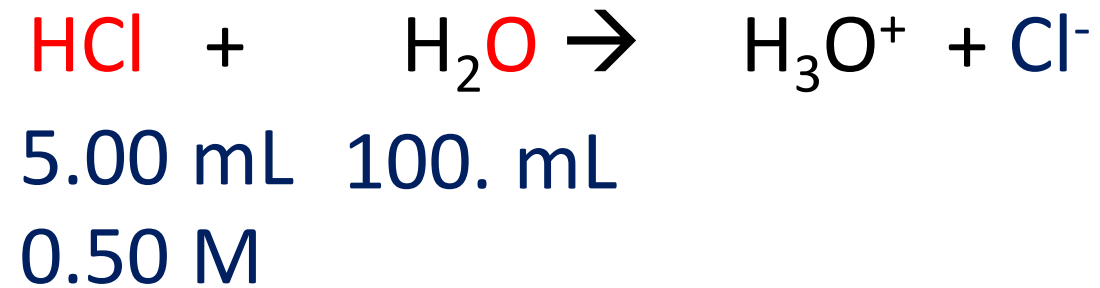
0.015 mol

	$\text{HC}_7\text{H}_5\text{O}_2$	$\text{C}_7\text{H}_5\text{O}_2^-$	OH^-
Before reaction	0.0050 mol	0.015 mol	0.0025 mol
Change	- 0.0025 mol	+ 0.0025 mol	- 0.0025 mol
After reaction	0.0025 mol	0.0175 mol	0.000 mol

$$\text{pH} = \text{pK}_a + \log \frac{[\text{C}_7\text{H}_5\text{O}_2^-] + [\text{OH}^-]}{[\text{HC}_7\text{H}_5\text{O}_2] - [\text{OH}^-]} \rightarrow \text{pH} = 4.19 + \log \frac{(0.0175)}{(0.0025)}$$

$$\text{pH} = 4.19 + 0.85 = 5.04$$

d. Calculate the pH when 5.00 mL of 0.50 M HCl is added to 100.00 mL of pure H₂O.



$$\begin{aligned} \text{Mole of HCl added} &= M \times V = 0.50 \text{ mol L}^{-1} \times (5.0 \times 10^{-3}) \text{ L} \\ &= 0.0025 \text{ mol} \end{aligned}$$

$$\text{Molarity, } [\text{H}_3\text{O}^+] = n/V_t = 0.0025/(105 \times 10^{-3})$$

$$\begin{aligned} \text{Molarity, } [\text{H}_3\text{O}^+] &= 0.0238 \text{ M} \quad \Rightarrow \quad \text{pH} = -\log(0.0238) \\ &\quad \boxed{\text{pH} = 1.62} \end{aligned}$$